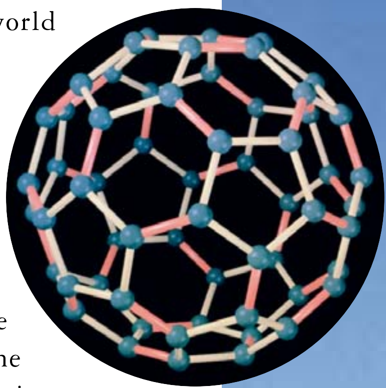
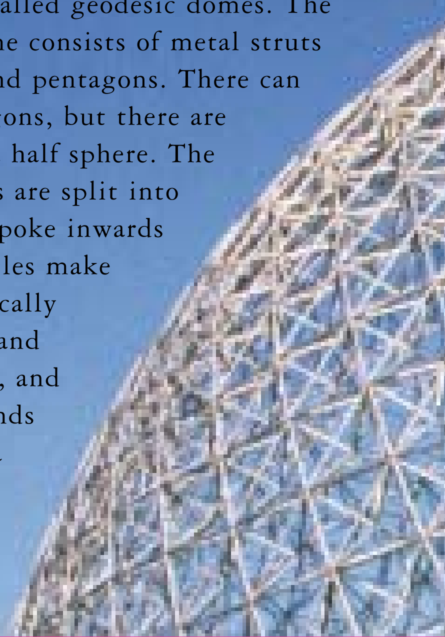
Footballs and *buckyballs*

If you chop all the corners off an icosahedron, you end up with a shape made of 20 hexagons and 12 pentagons, just like a football. It's called a "truncated icosahedron". (Footballs always have 12 pentagons, but the other panels can vary in shape and number). In 1985, three scientists amazed the world when they discovered a chemical with exactly the same shape. Each molecule is a cage of 60 carbon atoms arranged in pentagons and hexagons. The discoverers called it the "buckyball" and won the Nobel Prize for finding it.





Dome homes

Truncated icosahedrons form the basic plan of superstrong buildings called geodesic domes. The frame of a geodesic dome consists of metal struts arranged in hexagons and pentagons. There can be any number of hexagons, but there are always 6 pentagons in a half sphere. The hexagons and pentagons are split into triangles, which either poke inwards or outwards. The triangles make geodesic domes fantastically tough. They can withstand earthquakes, hurricanes, and burial under huge mounds of snow. There is even a geodesic dome on the South Pole, which has the worst weather on Earth.

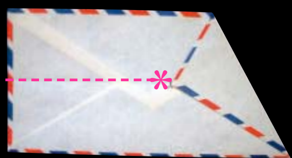
Things to do

Here's an easy way to make a **TETRAHEDRON** from an envelope:

1

Seal the envelope and fold it along the middle to make a faint crease.

2

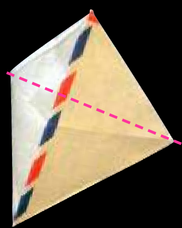


Fold a corner to the centre crease and make a mark with a pen where it meets.

3

Cut across the mark. Then make two strong creases between the mark and the corners and fold them both ways.

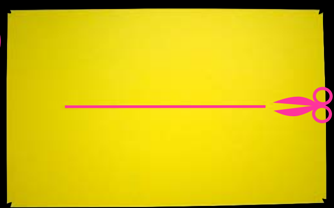
4



Open out the tetrahedron and tape the hole shut.

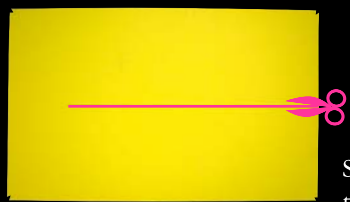
To make an **ICOSAHEDRON,** cut out 3 pieces of stiff card, each 13 by 21 cm (5.1 by 8.25 in) in size. Make small notches in all the corners.

1



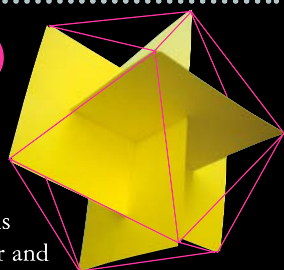
In each card, cut a slot just over 13 cm (5.1 in) long in the middle. Ask an adult to help.

2



In one of the cards, extend the slot all the way to one side.

3



Slot the cards together and wind thread between the corners to make an icosahedron.